

Part A – Theory: Regression in Machine Learning

❖ Introduction to Regression

Regression is a supervised machine learning technique used to predict continuous numerical values. It learns the relationship between input variables (features) and a target variable. Unlike classification, which predicts categories such as spam or not spam, regression predicts numbers such as car prices or house values. A real-life regression example is predicting the selling price of a used car. A classification example is predicting whether a customer will buy a product (Yes/No).

Types of Regression

- ❖ Linear Regression models a straight-line relationship between one feature and a target. It is simple, fast, and easy to interpret but cannot capture complex patterns.
- ❖ Multiple Linear Regression extends this idea by using multiple input features, making it suitable for predicting car prices from mileage, year, and other attributes.
- ❖ Polynomial Regression adds polynomial terms to model curved relationships. It can improve accuracy for nonlinear data but is more likely to overfit if the degree is too high.

Regression Metrics

Metric	Measures	Units	Large Errors
MAE	Average absolute error	Same as target	Low sensitivity
MSE	Average squared error	Squared units	High sensitivity
RMSE	Square root of MSE	Same as target	High sensitivity
R^2	Explained variance	No unit	Not an error metric

Underfitting and Overfitting

- ❖ Underfitting happens when a model is too simple to learn the relationship in the data.
- ❖ Overfitting occurs when a model memorizes the training data and performs poorly on unseen data. Polynomial Regression with a very high degree can easily overfit by fitting noise instead of the true pattern. Common ways to reduce overfitting include using simpler models, cross-validation, and limiting model complexity.

Real-World Case Study

A transportation company can use Multiple Linear Regression to estimate used-car prices. The dataset may include vehicle age, mileage, number of doors, accident history, and location. The regression model predicts selling prices to support fair pricing and business decisions. The results help dealerships estimate market value and improve pricing consistency.

References

- Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning.